

CRS-R State-of-Consciousness Transitions: Quick Reference

A complementary way to interpret serial CRS-R assessments: identify the highest-level behaviors demonstrated and determine whether they support a transition in state of consciousness.

State-of-consciousness continuum



Increasing evidence of wakefulness and/or conscious awareness →

How state-transition interpretation works

- 1. Examine the CRS-R subscale findings, not only the total score.** Diagnostic classification is driven by specific behaviors that provide evidence of consciousness.
- 2. Identify the highest-level behavior demonstrated.** Use the behavior-to-state mapping at right to determine whether the assessment supports UWS, MCS–, MCS+, or emergence from MCS.
- 3. Compare serial assessments.** A newly demonstrated behavior may move the patient across a diagnostic boundary even when the total-score difference is small.
- 4. Interpret state transition and measured change together.** A patient can improve substantially within the same state, while a strategically important behavior can produce a state transition with relatively little total-score change.

Key distinctions

UWS: CRS-R findings do not demonstrate the behaviors listed as MCS–, MCS+, or eMCS in the mapping table.

MCS–: At least one MCS– behavior is demonstrated, without an MCS+ or eMCS behavior.

MCS+: At least one MCS+ behavior is demonstrated, without an eMCS behavior.

eMCS: Functional object use or functional accurate communication is demonstrated.

Coma: This table from the Better Measurement paper maps CRS-R subscale behaviors from UWS through eMCS; it does not provide a separate CRS-R behavior column for coma. Coma therefore requires interpretation of wakefulness/arousal in addition to the behavior mapping shown here.

Important: State classification and change beyond measurement error answer different questions. Use both when interpreting serial CRS-R assessments.

CRS-R behaviors mapped to state of consciousness

The table below reproduces the behavior-to-state mapping presented in Table 2 of the supplied Better Measurement paper.

Subscale	eMCS	MCS+	MCS–	UWS
Auditory function	—	4 Consistent movement to command	3 Reproducible movement to command 2 Localization to sound	1 Auditory startle 0 None
Visual function	—	5 Object recognition	4 Object localization: reaching 3 Visual pursuit 2 Fixation	1 Visual startle 0 None
Motor function	6 Functional object use	—	5 Automatic motor response 4 Object manipulation 3 Localization to noxious stimulation	2 Flexion withdrawal 1 Abnormal posturing 0 None
Oromotor / verbal	—	3 Intelligible verbalization	—	2 Vocalization / oral movement 1 Oral reflexive movement 0 None
Communication	2 Functional: accurate	1 Nonfunctional: intentional	—	0 None
Arousal	—	—	—	—

Read across the row: each behavior is placed under the state category it supports. Classification is based on the highest diagnostic category supported by the behaviors demonstrated.